

Machine Learning Report: Regression

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Report Task 1: Model Summaries

Code Task 1: Linear Regression

The predicted against actual plot shows a generally linear trend, although there is noticeable variance around the perfect prediction line $y = x$.

Learned Coefficients The learned coefficients (rounded to 3 d.p.) are:

Feature	Coefficient
age	3614.975
bmi	2036.228
children	516.890
sex_female	9.296
sex_male	-9.296
smoker_no	-11825.564
smoker_yes	11825.564
region_northeast	459.585
region_northwest	88.908
region_southeast	-198.279
region_southwest	-350.214

Table 1: Linear Regression Coefficients

Model Performance Root Mean Squared Error (RMSE) & R^2 score:

- Training RMSE: **6105.545**
- Test RMSE: **5796.285**
- Training R^2 : **0.742**
- Test R^2 : **0.784**

Linear Regression Plot Figure 1 shows the plot for the linear regression model.

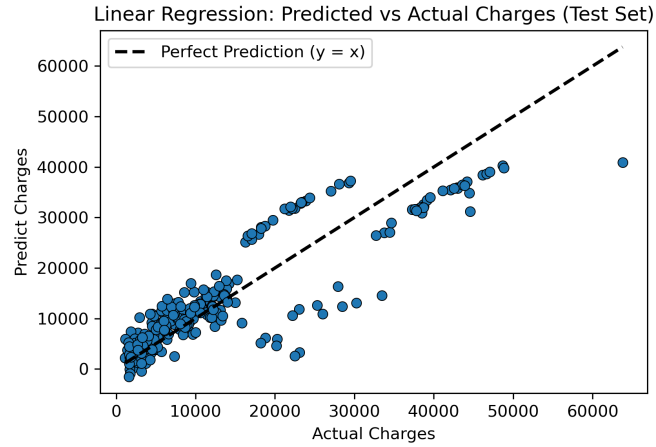


Figure 1: Linear Regression model.

Code Task 2: Neural Network

The Neural Network achieves good performance, with RMSE and R^2 values better than Linear Regression, indicating it can capture some non-linear relationship in the data.

Model Performance Root Mean Squared Error (RMSE) & R^2 score:

- Training RMSE: **4768.954**
- Test RMSE: **4524.837**
- Training R^2 : **0.814**
- Test R^2 : **0.856**

Neural Network Plot Figure 2 shows the plot for the neural network model.

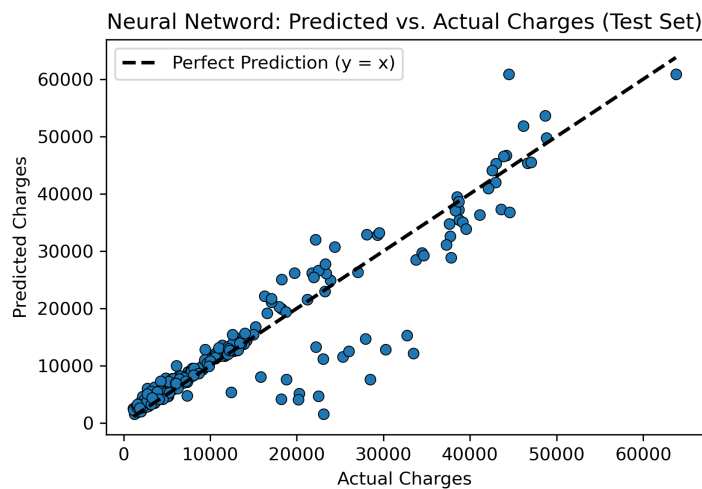


Figure 2: Neural Network model.

Code Task 3: Bayesian Linear Regression

The Bayesian Linear Regression provides uncertainty estimates, offering a more interpretable probabilistic view of the regression than Linear Regression and Neural Network.

Posterior Means The posterior means (rounded to 3 d.p.) are:

Coefficient	Posterior Mean
w_0 (intercept)	911.162
σ (noise std.)	2305.211
age	250.615
bmi	174.416
children	63.207
sex_female	418.804
sex_male	492.681
smoker_no	419.488
smoker_yes	492.045
region_northeast	228.140
region_northwest	205.163
region_southeast	267.299
region_southwest	210.267

Table 2: Posterior mean coefficients of the Bayesian linear regression model.

Trace Plot All chains mixed well and their $\hat{R}$ scores are extremely close to 1.00, indicating the MCMC has converged

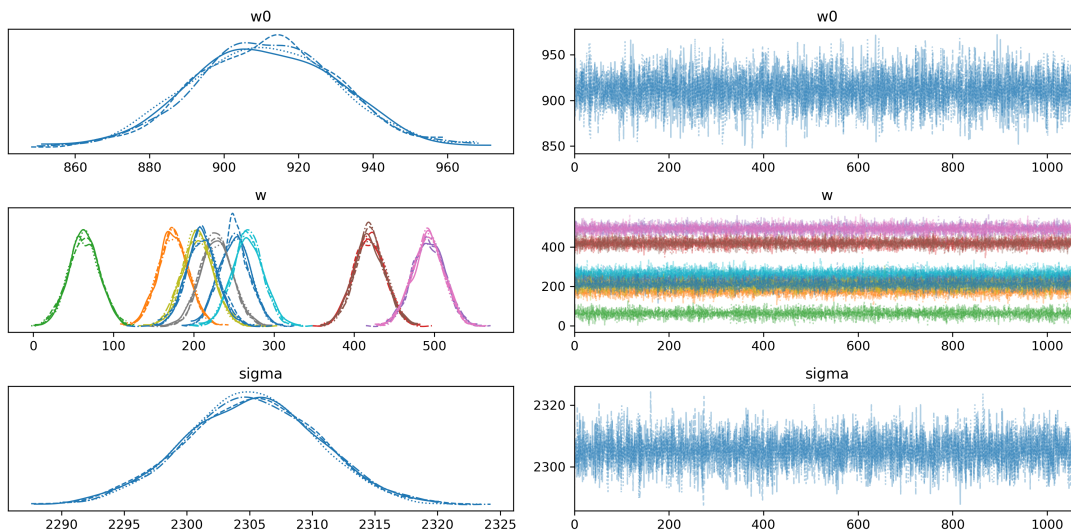


Figure 3: Trace plot for parameters

Report Task 2: Model Comparison and Discussion

Test RMSE Comparison

The difference in RMSE reflected the average size of the prediction error for each model. The Linear Regression model achieved a test RMSE of **5796.285**, while the Neural Network obtained a test RMSE of **4524.837**. This indicates that the **Neural Network** has a lower average prediction error on unseen data.

Model Complexity and Interpretability

Linear Regression is a simple and highly interpretable model, where each learned coefficient directly describes the effect of its corresponding feature on the predicted insurance charges.

In contrast, the Neural Network is a more complex model with multiple hidden layers and non-linear activations. Although it can represent more flexible relationships, it is substantially less interpretable, as individual feature effects cannot be easily extracted from the network parameters.

Which Model Performs Better and Why

Based on the test RMSE values, the Neural Network performs better. The Neural Network achieves a lower RMSE, its improved performance is due to its ability to model the non-linear relationship between the features that Linear Regression cannot capture.

The $\hat{R}$ score measures the proportion of variance in the target variable that is explained by the model. The Neural Network $\hat{R}$ score (0.856) is higher than the Linear Regression $\hat{R}$ score (0.784). This suggests the Neural Network captures more variability in the data and a better fit than Linear Regression.